Native gates or nothing: the condition for a qudit advantage in uncorrected quantum algorithms under decoherence

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Whether qudits buy resilience against decoherence is a recognized open question: prior work compares algorithms by noiseless resource counts, or error-correcting codes, which have no problem instance to compress. We simulate Shor order finding, eigenstate phase estimation, and Grover search as bare, uncorrected circuits—the regime of near-term demonstrations—on qubit, qutrit, and ququint registers (with a demo-size $d = 7$ check) under two decoherence channels—a ladder channel calibrated to published per-level transmon coherence data and a depolarizing channel representative of trapped-ion qudits—three entangling-gate cost models, and registers to 3.9×10^5 Hilbert dimensions (18.6 qubit-equivalents). One condition governs all three algorithms: *qudits outperform qubits only with a native two-qudit entangling gate whose cost grows no faster than linearly in d.* Gates compiled by two-level decomposition forfeit the advantage in every case tested but one—Shor at $d = 3$ under per-particle noise, where width compression alone survives the depth surcharge; whether linear cost suffices is set by the level and structure of the operating dephasing—the advantage holds on per-particle and refocused-ladder hardware, free-evolution ladder dephasing splits the linear-cost cells, and unmitigated Zeeman-structured dephasing reverses the verdict outright. Along the way we identify a number-theoretic confound—grid alignment of the phases s/r—that reverses naive cross-dimension Shor comparisons and predicts the winner in every biased instance tested (three instances $\times$ two noise models), and we reduce the mechanism to a quantitative law: accumulated channel damage fixes end-state fidelity on a single exponential across algorithms and bases, with amplitude $\approx$1 and $R^2 = 0.97$–0.99 (the first-order expectation; in log fidelity the law holds within a factor of two until states approach their 1/dim floor); the algorithm enters only through its decoder's error tolerance, which we reduce to an exact number-theoretic law verified outcome-for-outcome against the decoder. The qubit branch of our predictions is anchored on hardware: the shallow compiled circuit reproduces its predicted success band on a commercial trapped-ion processor (0.617 ± 0.007 vs 0.60–0.70), pinning the device's effective depolarizing strength at its measured per-gate infidelity.

I. INTRODUCTION

Quantum hardware is not binary by nature. Transmons, trapped ions, nitrogen-vacancy centers, and molecular spins all expose more than two usable levels, and processors that exploit them as qudits now exist on several platforms [1–6]. The theoretical case for higher dimensions is old and, at the level of structure, strong: fault-tolerant stabilizer constructions close cleanly in prime dimension [7] and the discrete phase-space structure is cleanest in odd prime dimension [8]; magic-state distillation protocols exist in every prime dimension [9], with thresholds and efficiencies that keep improving with d (within congruence classes mod 3, for primes up to 17) [10]; and qudit memory thresholds rise under abstract noise models [11]. A base-d register also compresses algorithms: at matched problem size it needs $\log_2 d$ times fewer carriers and a proportionally shorter schedule.

Whether any of this survives *physical* decoherence—in which higher levels of a real device decay and dephase faster than lower ones—is a question the literature has flagged repeatedly and not answered. Campbell noted that "in physical systems one may also see noise rise with d. Such features depend subtly on the details of the underlying physics" [10]; Marks *et al.* point to the "increased degrees of freedom that can be coupled to the environment" and call whether such noise can be kept sufficiently small "an interesting question for experimental implementations" [11]; Pavlidis and Floratos, citing qubit evidence that order finding tolerates truncation of small QFT rotations [12, 13], conjecture "a similar robustness" for qudits but leave its investigation open [14]; and the recent qudit review of Kiktenko *et al.* lists "the investigation of the impact of noise within the discussed schemes" as an open problem [15]. Comparisons *across* dimensions still treat noise indirectly: entangling-gate counts as a noise proxy [15, 16], analytic coherent-error bounds [17], or the qualitative argument that fewer carriers means less local noise [18]. Noisy qutrit circuit simulations exist [19, 20], but always at $d = 3$ against a qubit baseline; no study has compared several prime dimensions on full algorithms under channels calibrated to measured per-level coherences, or followed such a comparison to scale. A claim that qudit phase-estimation error decreases exponentially with d—attributed by the review of Wang *et al.* [18] to a conference contribution of Parasa and Perkowski [21], whose slides state it for the algorithm's intrinsic failure probability at fixed precision (the full text we were unable to obtain)—concerns

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the noiseless algorithm and has, to our knowledge, never been tested against a noise model.

This paper answers the question quantitatively at the algorithm level. We simulate Shor order finding [22], eigenstate quantum phase estimation (QPE) [23], and Grover search [24] on qubit ($d = 2$), qutrit ($d = 3$), and ququint ($d = 5$) registers, under two decoherence channels chosen to bracket real hardware—an anharmonic-ladder channel calibrated to published per-level transmon coherence measurements [2, 3, 25–27], and a per-particle depolarizing channel representative of trapped-ion qudits [1, 4]—under three entangling-gate cost models spanning native gates, the measured trapped-ion $2(d{-}1)$ Mølmer–Sørensen cost [1], and $O(d^2)$ two-level decomposition [14], with registers to 3.9×10^5 Hilbert dimensions (18.6 qubit-equivalents; the widest qubit register is 17 qubits) reached by quantum trajectories; a demo-size $d = 7$ run (Sec. IV) extends the cost grid one prime higher.

Our central result is a condition rather than a verdict:

> *Qudit gates compiled by two-level decomposition ($O(d^2)$ layer cost) forfeit the advantage in every algorithm, dimension, and channel we test, with a single exception—Shor at $d = 3$ under per-particle noise, where width compression alone survives the depth surcharge (Sec. IV). A native entangling gate whose cost grows no faster than linearly in d delivers the advantage when the operating dephasing is weak or unstructured: on per-particle hardware and on refocused ladder hardware. Free-evolution ladder dephasing splits the linear-cost cells by algorithm and dimension, and unmitigated Zeeman-structured dephasing—higher levels dephasing tens of times faster—reverses the verdict at every cost model (Secs. IV and VIII). Quantitatively, the break-even is the qudit's layer-count ratio clearing $(d^2 - 1)/(3 \log_2 d)$ at the operating dephasing level.*

The condition is a statement about bare, uncorrected circuits—the regime of every near-term demonstration. It does not transfer automatically to the error-correction layer, where a code has no problem instance to compress and the dimension dependence can carry the opposite sign [28] (Sec. X).

The native-gate requirement is met by current trapped-ion qudit gates and by native cross-Kerr transmon entanglers [3], and violated whenever qudit gates are compiled by two-level decomposition [14, 29]. It explains the split verdicts in prior work: qutrit advantages obtained with native third levels—as ancillas [19] or as encoding capacity [20]—versus 35–69% more non-Clifford gates for single-qutrit synthesis from a fault-tolerant gate set [29]. An independent analytic gate-level criterion [30], which we reproduce numerically to 4×10^{-4} and apply to our circuits, predicts our algorithm-level outcomes in seven of nine cost-model/dimension cases (both misses in the conservative direction) and sets the break-even gate-efficiency ratio at only 1.68 ($d = 3$), 3.45 ($d = 5$), and 5.70 ($d = 7$)—far below the $O(d^2/\log_2 d)$ folklore values of 5.7, 10.8, and 17.5.

Three further results are methodological and, we believe, of independent use to the field.

(i) Grid alignment is a confound in every cross-dimension comparison of order finding (Sec. III). Phase estimation concentrates probability on the phases s/r; when the order r divides the control dimension $D = d^m$ these sit exactly on measurement grid points and the peaks are maximally noise-robust. Which base receives this gift is decided by number theory, not physics. The canonical demonstration instance $N = 15$ admits *only* power-of-two orders, silently handing qubits perfect alignment; on unbiased instances the ordering of bases reverses. Alignment predicts the winner in all six biased runs we construct (three aligned instances × two noise models), and a within-modulus control—two orders hosted on identical registers—prices it at ≈0.2 in floor-corrected signal.

(ii) The mechanism is width-and-depth compression, and it reduces to a quantitative law with a stated boundary (Secs. VI–VII). Grover search holds the oracle count fixed across bases, delivering roughly half of Shor's exposure compression: the qudit advantage survives (confirming a prediction recorded in the source before the runs) at 0.33–0.50 of Shor's. Counted in *damage units*—the per-carrier-layer entanglement infidelity $1 - \operatorname{tr} S/d^2$ of the actual channel—accumulated exposure places end-state fidelity for both algorithm families on a single exponential ($R^2 = 0.97$–0.99, amplitude ≈1). What exposure does not determine is the measured success probability, because the classical decoder intervenes: continued-fraction order recovery gains error tolerance with register size, and we observe it decoding correct orders from states with fidelity 3×10^{-4} to the ideal state. That tolerance itself obeys an exact law—a totient sum over the denominators the decoder admits—which we derive and verify outcome-for-outcome against the decoder (Sec. VII C).

(iii) Standing reviewer objections are pre-empted with data (Sec. VIII): d-dependent readout error is structurally near-neutral (the growth of per-level misread rates with d cancels against the reduced carrier count), and refocused operation helps qudits *more* than qubits in all conditions tested, moving the break-even point of the central condition by roughly one cost model. The same section stress-tests the noise conventions themselves: against inflated per-layer qudit noise, against Zeeman-structured dephasing (where the condition genuinely fails unmitigated), and against composite dimensions $d = 4$ and 6.

Finally, the qubit branch of the predictions is anchored on real hardware (Sec. IX): on a commercial trapped-ion processor the shallow compiled circuit lands inside its predicted band—validating the depolarizing convention on the platform it models—while the deep-circuit and fixed-connectivity failures delineate, on hardware, the coherent-systematic and compilation-overhead regimes our channels deliberately exclude.

Prior work closest to ours is differentiated in Sec. X: the gate-level criterion of Janković *et al.* [30] (no algorithms, no scaling), the qutrit circuit studies of Gokhale *et al.* [19] and Gustafson [20] ($d = 3$ only; no comparison across multiple prime dimensions, no per-level-calibrated channel), and the code-level comparison of Keppens *et al.* [28], whose opposite ordering we reconcile explicitly: a code has no problem instance to compress, so it pays the per-carrier cost of dimension without the width-and-depth rebate that is the entire source of the algorithmic advantage measured here.

II. SETUP

Figure 1 summarizes the simulation pipeline and the exposure accounting that every result in this paper shares; the subsections below specify each stage.

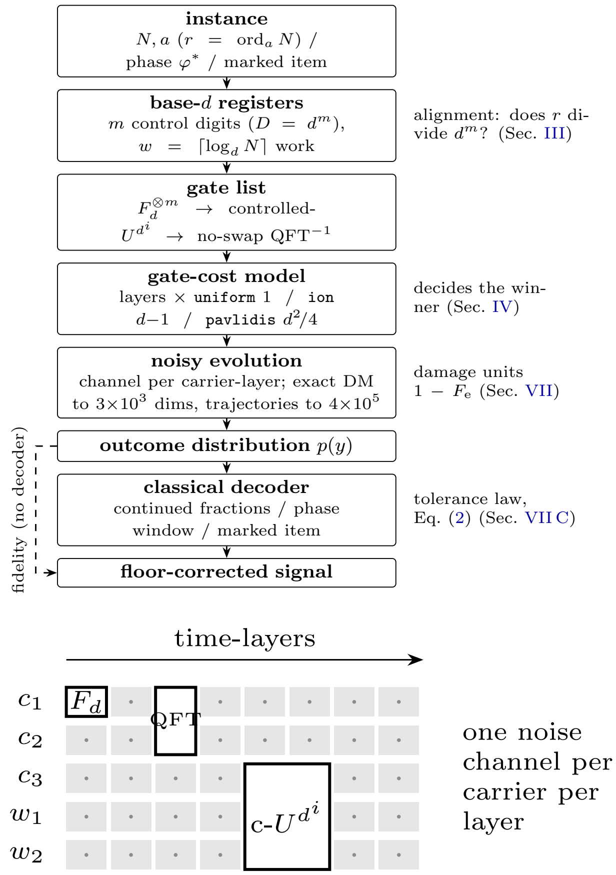

FIG. 1. Top: the simulation pipeline. Every algorithm, base, and channel flows through the same stages; the four annotations mark where the paper's main objects enter—the alignment confound at register choice, the cost model that decides the winner, damage-weighted exposure in the noise stage, and the decoder whose tolerance obeys Eq. (2). The dashed path is end-state fidelity, which bypasses the decoder and obeys the one-curve damage law of Sec. VII. Bottom: the exposure convention. Rows are carriers (control c_i, work w_i), columns are time-layers; every carrier passes through the noise channel every layer (dots), a gate spanning k carriers occupies its cost-model depth in layers (boxes), and idling carriers decohere while gates execute. Exposure = carriers × layers; in damage units it is exposure $\times(1 - F_e)$.

A. Algorithms and registers

We compare three algorithms chosen to span the failure modes of interference under decoherence. *Shor order finding*: given N and a coprime a, find the multiplicative order r of a mod N, using a control register of m base-d digits (control dimension $D = d^m$) and a work register of $w = \lceil \log_d N \rceil$ digits; success is the probability that continued-fraction post-processing of the measured outcome recovers r exactly. Our benchmark instances are unbiased in the sense of Sec. III: $N = 21$, $a = 2$ ($r = 6$) and $N = 29$, $a = 16$ ($r = 7$). *Eigenstate QPE*: phase estimation on an eigenstate of a diagonal unitary with target phase equal to the golden-ratio conjugate $\varphi^* \approx 0.6180$, chosen to be far from every fraction with a small base-2, -3, or -5 denominator; success is the estimate landing within $2^{-(b+1)}$ of the target ($b = 5$ bits). *Grover search* over $M = d^n$ items with $\lfloor(\pi/4)\sqrt{M}\rceil$ iterations; success is the probability of measuring the marked item, averaged over a sample of marked items excluding $|0\cdots0\rangle$ (the diffuser's own reflection axis and the bottom of the damping ladder).

Because $M = d^n$ never matches exactly across bases, all cross-base comparisons are made at matched problem size by interpolating each base's results onto a common $\log_2$-size axis (linear interpolation of the signal in $\log_2$ size); comparing raw demo-size points reverses orderings and is one of the two fairness traps we document (the other is alignment).

B. Noise channels

One time-layer of noise is applied to *every* carrier for every layer of the serial schedule; a gate spanning k carriers occupies its decomposition depth in layers (below), so idling carriers decohere while gates execute.

Calibrated ladder (transmon-like). A Lindblad channel with sequential relaxation $\Gamma_k \propto k^{0.7}$ and pure dephasing following a max-level law $\Gamma_\phi(j,k) \propto \max(j,k)^{1.1}$, exponentiated exactly per layer. Both exponents are fits to published per-level coherence data spanning nine devices and $d = 3$ to 12: the relaxation ratio Γ_2/Γ_1 is measured at $\approx$1.7 [2, 3, 25–27, 31] against 2.0 for the textbook $\Gamma_k \propto k$ ladder, and the dephasing ratios $\Gamma_\phi^{01} : \Gamma_\phi^{12} : \Gamma_\phi^{02}$ are measured at 1 : 2.0 : 2.3—incompatible with the textbook $(\Delta\text{level})^2$ law (which predicts 1 : 1 : 4) because the charge dispersion of $|2\rangle$ exceeds that of $|1\rangle$ by an order of magnitude [2]. The max-level dephasing target is realized *exactly* by diagonal jump operators obtained through classical multidimensional scaling (Sec. XI); our channel reproduces the measured ratios as 1.62 and 1 : 2.14 : 2.14. Rates are normalized so the $0\leftrightarrow1$ subspace is bit-for-bit identical to the qubit channel: every difference between bases is purely a higher-level effect. A dephasing-scale knob interpolates to the high-E_J/E_C regime demonstrated at $d=12$ [27], where echo coherence approaches the T_1 limit; the same knob models refocused operation (Sec. VIII).

Per-particle depolarizing (ion-like). $\rho \to (1-p)\rho + p\,\mathbb{I}/d$ per carrier-layer. The convention rests on measured trapped-ion structure: every qudit level is ground-state or metastable ($\tau_1 \sim 1.1\,\mathrm{s}$, orders of magnitude beyond gate times), allowed-transition magnetic sensitivities span only a factor of $\sim$5 (pair dephasing *rates* go as the square, and idling pairs beyond the driven set widen the span—Sec. VIII prices the full anisotropy, a 1–49× rate spread at $d=5$), and the single-qudit per-pulse error is nearly flat in d (2.0×10^{-4} at $d=3$, 3.2×10^{-4} at $d=5$) [1], with qudit control demonstrated to 13 levels [4]. A Zeeman-structured dephasing variant that keeps the encoding's full pair anisotropy is tested in Sec. VIII.

Measured per-gate noise strengths (rate × gate time) place transmon two-qutrit gates at 7×10^{-3}–3×10^{-2} [3] and ion two-qudit gates at $\sim10^{-3}$ [1, 4]; our sweeps (0.001–0.05 at demo size, 0.003–0.005 for scaling) bracket both operating points, which are marked on every figure.

Four named noise regimes appear in the figures: the *idealized ladder* ($\Gamma_k \propto k$, $(\Delta\text{level})^2$ dephasing—the textbook model, used only as a bound), the *calibrated ladder* above, its *low-charge-dispersion* variant (the dephasing knob set to the high-E_J/E_C regime of Ref. [27]), and *per-particle depolarizing.*

C. Gate-cost models

Real hardware charges more per entangling gate as d grows, and this is the strongest objection to any qudit advantage. We charge three published cost structures as layer multipliers: `uniform` (native qudit entangler, one layer regardless of d—e.g. the cross-Kerr CZ of Ref. [3]); `ion` ($d-1$ per entangling gate, Ringbauer's $2(d-1)$ Mølmer–Sørensen construction normalized to 1 at $d=2$—the d=2 MS gate is itself the two-pulse unit, so the model charges relative pulse count [1]); and `pavlidis` ($d^2/4$ on all gates, the harshest of the three: the controlled rotations of the QFT-arithmetic circuits of Ref. [14] decompose into $4(d-1)^2$ elementary two-level gates, i.e. $(d-1)^2$ per gate after the same $d=2$ normalization, and our $d^2/4$ charge rounds that cost *down*—2.25 against 4 at $d=3$, 6.25 against 16 at $d=5$—so the verdicts against decomposed gates below are conservative). The three models swing the ququint Shor circuit from 3.8× *shorter* than the qubit's (15 vs 57 layers on $N=21$) to 1.6× *longer* (93.8 vs 57)—and that swing, not the noise model, decides the winner. Multi-qudit gates in Grover (oracle and reflection) are applied as exact unitaries but charged their $n-1$-layer decomposition depth; charging them one layer would hand the largest free ride to the base packing the most carriers into the gate, which is $d=2$. Unless a figure states otherwise, plotted results use the `uniform` (native-entangler) cost model—the most qudit-favorable of the three; Table I and Secs. IV, VI, and VIII quantify how the winner changes under `ion` and `pavlidis`.

D. Success metric

Raw success probabilities have base- and instance-dependent random floors: a uniformly random outcome passes continued fractions with probability 0.12–0.13 on the $N=21$ benchmark at demo size, 0.20–0.26 on $N=29$, and up to 0.59 on small-order instances. All comparisons therefore use the floor-corrected signal

$$\text{signal} = \frac{\text{success} - \text{floor}}{\text{success}_{\text{noiseless}} - \text{floor}}, \tag{1}$$

which is 1 for perfect interference, 0 for random guessing, and negative when noise actively biases outcomes away from the answer. Floors and noiseless baselines are recomputed for every base, size, and readout-error setting. Because small orders compress the metric's dynamic range, we require a floor-to-baseline span above 0.15 for any instance used in a quantitative comparison; the one exception is the $r=3$ aligned instance of Fig. 2 (spans 0.07–0.12), retained because no other instance represents its alignment class—its winner margins far exceed its metric compression, but we quote no signal magnitudes from it.

III. GRID ALIGNMENT: A CONFOUND THAT DECIDES WINNERS

Phase estimation concentrates probability on the phases s/r. If r divides $D=d^m$, these sit exactly on grid points of the base-d measurement basis and the interference peaks are perfectly sharp; otherwise they smear over neighboring outcomes, and smeared peaks are far more fragile under noise. Which base receives sharp peaks is decided by the arithmetic of the instance, not the physics of the hardware—and it is invisible in the existing literature because prior studies work at fixed d, where alignment is a constant.

The canonical instance $N=15$ is maximally pathological: its multiplicative group is $\mathbb{Z}_2\times\mathbb{Z}_4$, so *every* order is a power of two and the qubit register always lands exactly on grid while $d=3,5$ never do. Any cross-dimension comparison run on $N=15$—including the first version of this study—hands base 2 a structural gift unrelated to decoherence.

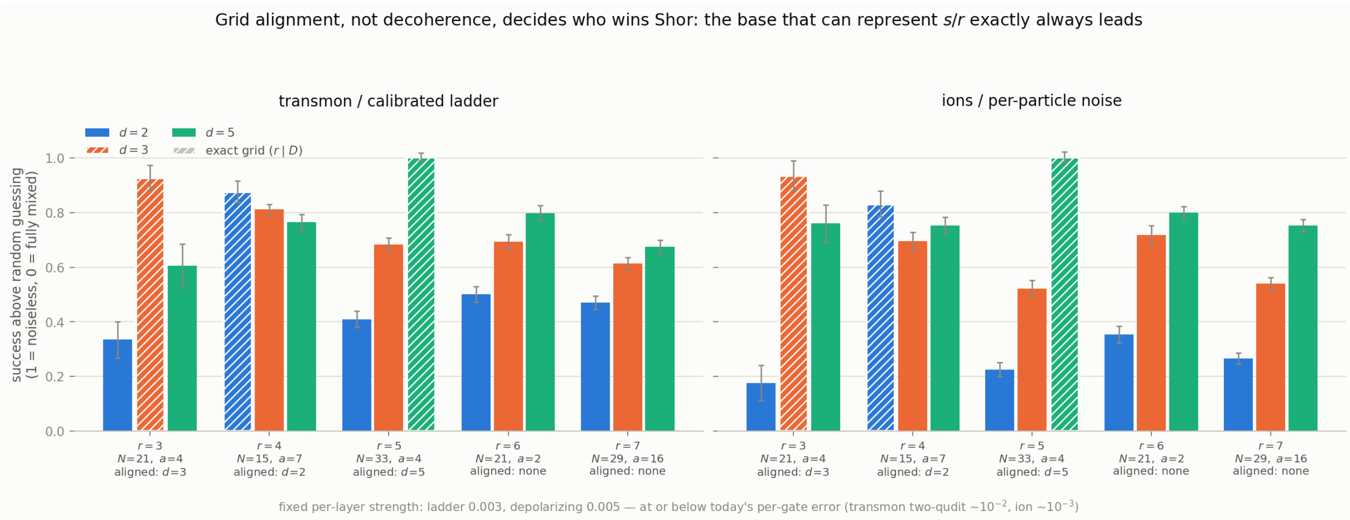

FIG. 2. Floor-corrected Shor signal for one instance per alignment class ($r = 3, 4, 5, 6, 7$), under calibrated-ladder (strength 0.003) and per-particle (strength 0.005) noise, `uniform` cost model. The base for which r divides d^m wins in all six biased runs (three aligned instances $\times$ two noise models); on both unbiased instances the ordering is $d{=}5 > d{=}3 > d{=}2$.

Figure 2 shows one instance per alignment class. The aligned base wins in **six of six** biased runs, with margins exceeding anything decoherence produces between bases (at $r = 5$ the aligned ququint scores 1.00 against the qubit's 0.23–0.41). On the two unbiased instances ($r = 6$, $r = 7$) all four runs give the strict ordering $d = 5 > d = 3 > d = 2$. Misalignment is moreover graded, not binary: averaging the distance of the phases s/r to the nearest grid point, the $r = 6$ instance is mildly tilted *toward* the qubit (0.267 vs 0.300) and the qubit still loses by 0.19–0.45, while $r = 7$ ($N = 29$) is exactly neutral across all three bases—the cleanest instance we know of, and the one we recommend for future benchmarks.

Within-modulus control. Varying N also varies register widths, so we isolate alignment with two moduli that each host an aligned and an unaligned order on *identical* registers: $N = 33$ and $N = 55$ carry both $r = 5$ (exact in base 5) and $r = 10$ (exact in no base) with the same (m, w), gate counts, and noise exposure per base. Removing alignment costs the ququint 0.14–0.22 signal and narrows its lead over the qubit by 0.20–0.26, consistently across both moduli and both noise models—while the ququint's *residual* lead on the unaligned orders remains 0.38–0.57, twice the alignment term. Both effects are real; alignment is worth ≈0.2 signal to whoever receives it, and the remaining qudit lead is physical. (A converse base-2 control is impossible to construct: at every modulus large enough to carry both $r = 4$ and a non-power-of-two order, the $r = 4$ continued-fraction floor exceeds its noiseless baseline, collapsing the metric—$N = 15$ is the *only* qubit-aligned instance with usable dynamic range, which is worth knowing in itself. A consequence is that the ≈0.2 alignment price is measured only in the ququint-aligned direction and has no converse check.)

Ensemble over the multiplicative group. Shor draws a uniformly from the units of $\mathbb{Z}_N$, so we also score the full ensemble at $N = 21$: all $a \neq 1$ in $\mathbb{Z}_{21}^{\times}$ (eleven values; orders $r = 6, 3, 2$ with multiplicity $6, 2, 3$), exact density-matrix evolution, `uniform` cost, both marked operating points, on the demo registers $D = 64/81/125$. Every $r = 2$ and $r = 3$ instance falls below the 0.15 span rule of Sec. II in every base—small orders collapse the metric, as disclosed there—so the scorable ensemble is precisely the unbiased $r = 6$ class, where the ensemble-mean signal preserves the qudit ordering: 0.52/0.73/0.76 for $d = 2/3/5$ on the calibrated ladder and 0.33/0.67/0.79 under depolarizing. Raw success needs no span but carries base-dependent floors (0.38/0.36/0.39 on average here): subtracting them, the full-ensemble excess is −0.006/+0.030/+0.019 (ladder) and −0.006/+0.028/+0.025 (depolarizing)—so the ensemble-robust statement is that both qudits clear the qubit, while the $d{=}5 > d{=}3$ ordering is a property of the scorable class rather than of the full ensemble. The ensemble a Shor user actually samples thus reproduces the fixed-instance conclusions. The same test at $N = 33$ (all $a \neq 1$, nineteen values; orders $r = 10, 5, 2$ with multiplicity $12, 4, 3$; 400 trajectories per point) adds alignment gifts in both directions and closes the loop quantitatively. The $r = 2$ class—base-2 aligned—is unscorable in every base, floors meeting or exceeding baselines, the converse collapse documented above; the scorable ensemble splits into the base-5-aligned $r = 5$ class, where the ququint scores 1.00/0.97 (ladder/depolarizing), and the unbiased $r = 10$ class, where the ordering $d{=}5 > d{=}3 > d{=}2$ holds without any gift (0.82/0.60/0.45 and 0.78/0.51/0.23). The ququint's aligned-over-unaligned excess, +0.19/+0.18, independently reproduces the ≈0.2 alignment price measured by the within-modulus control: the ensemble structure is predicted class by class.

TABLE I. Floor-corrected signal on the unbiased instance ($N = 21$, $r = 6$) under three gate-cost models, at a single common strength of 0.005 for both channels (the middle of the sweep; ladder figures elsewhere use the marked transmon operating point 0.003). Rightmost column: total circuit cost in time-layers for $d = 2/3/5$. The winner is decided by the cost model, not the noise model.

noise	cost	d=2	d=3	d=5	layers
depol.	`uniform`	0.331	0.667	**0.782**	57/26/15
	`ion`	0.331	0.497	**0.502**	57/44/42
	`pavlidis`	0.331	**0.394**	0.215	57/58.5/93.8
ladder	`uniform`	0.282	0.578	**0.631**	57/26/15
	`ion`	0.282	**0.374**	0.256	57/44/42
	`pavlidis`	**0.282**	0.255	0.039	57/58.5/93.8

At $N = 55$ (all $a \neq 1$, thirty-nine values; adds the base-2-aligned $r = 4$ class and $r = 20$) the scorable pattern replicates—the ququint scores 1.00/0.99 on its aligned class, the unbiased $r = 10$ ordering is 0.82/0.58/0.44 and 0.81/0.49/0.23, the aligned excess is again +0.18, and every base-2-aligned class ($r = 2, 4$) is unscorable, the converse collapse now observed at all three moduli. The modal class, $r = 20$ (sixteen of the thirty-nine), is unscorable in *every* base, and for $d = 2$ maximally so: at $D = 64$ the noiseless baseline is exactly zero—the $r = 20$ acceptance window contains no outcome at all—the decoder law of Sec. VII C deciding scorability before noise enters.

Re-running the full noise sweep on the unbiased instance (Fig. 3) reverses the study's original conclusion: the best qudit beats the qubit at every noise strength under *all three* noise models—including the idealized ladder ($\Gamma_k \propto k$, $(\Delta\text{level})^2$ dephasing), the harshest treatment of high levels we ever used, under which the qubit now trails by 0.09–0.52. The earlier "qubits win Shor on transmons" finding was not a noise result at all; it was an arithmetic artifact, and it does not survive deconfounding under any model.

IV. THE COST CONDITION

Table I is the paper's central result in one table. One direction of the condition holds with a single, instructive exception: under d^2 decomposition (`pavlidis`) qubits win in every algorithm, dimension, and channel tested *except* Shor at $d = 3$ under per-particle noise (0.394 vs 0.331 in Table I, replicated at 1000-trajectory statistics in the $d = 7$ demo grid, 0.378 vs 0.312). The exception isolates the width component of the mechanism: at $d = 3$ the `pavlidis` depth surcharge almost exactly cancels the depth compression (58.5 vs 57 layers), while the width compression (7 vs 11 carriers) survives, and per-particle damage is nearly flat in d—so width alone carries a residual advantage that the ladder channel, whose per-event damage doubles at $d = 3$, erases. Everywhere else in the grid—the ladder column, the $d = 5$ rows, eigenstate QPE (a dead heat, 0.396 vs 0.395), and Grover in both channels—decomposition forfeits the advantage. Under a native qudit gate (`uniform`) the width-and-depth compression is untouched and qudits win on both hardware classes. The intermediate linear cost (`ion`) is where dephasing level decides: on per-particle hardware the ququint advantage survives (0.502 vs 0.331), while at free-evolution ladder dephasing the verdict splits—the qutrit still wins Shor but the ququint loses (0.256 vs 0.282), eigenstate QPE is a dead heat (advantage +0.00), and Grover passes to the qubit for both qudits (Sec. VI). Refocusing repairs this entire column (Sec. VIII). Eigenstate QPE (never confounded, its golden-ratio target phase being irrational in every base) otherwise tracks Shor, with advantages of +0.42 (`uniform`) and +0.20 (`ion`) on per-particle noise—so the decomposition direction holds for three genuinely different algorithms, while the converse is conditional on the dephasing level exactly as the central condition states.

Cost model and noise model are not independent: each pairing describes a platform, and both *physically matched* pairings favor ququints—trapped-ion qudits (`ion` cost, per-particle noise) and native-cross-Kerr transmons (`uniform` cost, calibrated ladder). The ion pairing presumes shielded, refocused operation; Sec. VIII prices the unmitigated alternative, and it is severe. The pessimistic cells of the grid are mostly mismatched combinations; the genuine failure mode is the absence of a native entangler (`pavlidis`), which is precisely the regime of synthesis-based compilation [29, 32].

Matched control dimension. The demo grid compares $D = 64/81/125$, and Sec. VII C shows decoder tolerance grows with D—the ququint enters Table I with twice the qubit's acceptance set (16 vs 8 outcomes). Equalizing D resolves the worry in the qudits' favor: at $d = 2$, $m = 7$ ($D = 128$, matching the ququint's 125) the qubit scores 0.47/0.30 (ladder/depolarizing) against 0.51/0.33 at $D = 64$—its extra decoder tolerance is outweighed by the deeper circuit's added exposure—while the ququint holds 0.76/0.78 and the qutrit is stable between $D = 81$ and 243 (0.73/0.67 vs 0.71/0.66). Unequal D biases the demo grid against the qudits, not for them.

An independent check: Janković *et al.* [30] derive, by linear response over Haar-random gates under pure dephasing, the critical gate-efficiency ratio $(d^2 - 1)/(3 \log_2 d)$ a qudit must clear to beat a qubit register—1.68 at $d = 3$, 3.45 at $d = 5$, and 5.70 at $d = 7$, far below the $O(d^2/\log_2 d)$ folklore. Reading our layer-count ratios as gate-efficiency ratios, their gate-level criterion predicts our algorithm-level winner in seven of nine cost/dimension cases on the ladder channel, including the tight case ($d = 5$ `uniform` clears 3.45 at 3.80 and wins). Both misses ($d = 3$ `ion`, and $d = 7$ `uniform` below) are in the conservative direction their pure-dephasing assumption predicts, our calibrated relaxation being gentler ($\Gamma_k \propto k^{0.7}$).

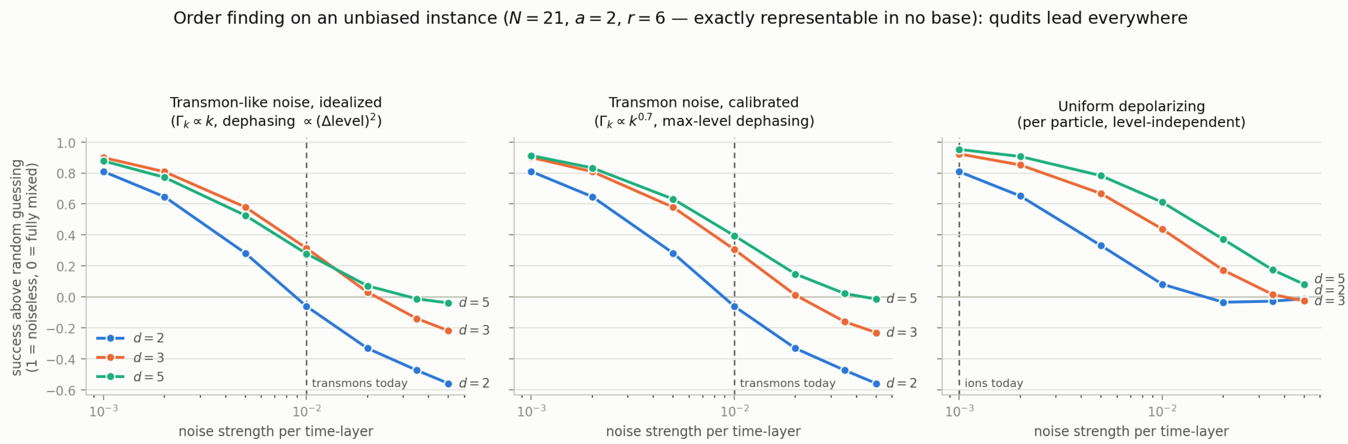

FIG. 3. Unbiased-instance noise sweep ($N = 21$, $a = 2$, $r = 6$), exact density-matrix evolution, `uniform` cost model (native entangler). Floor-corrected signal vs noise strength for the idealized ladder, calibrated ladder, and per-particle depolarizing channels. The best qudit leads the qubit at every strength under all three models; the single exception to a uniform qudit lead is the deepest depolarizing point ($s = 0.05$), where the qutrit alone dips marginally below the qubit with all signals within 0.08 of zero.

The seventh dimension. A demo-size $d = 7$ run on the unbiased instance (1000 trajectories per point, both channels, all three cost models; base 7 has the same mean residual misalignment, 0.300, as bases 3 and 5) extends the grid one prime higher and sharpens the condition. The decomposition direction is unchanged: `pavlidis` sits at or below the random floor (-0.05 to $+0.04$ signal). The native-gate direction survives—under `uniform` cost the $d = 7$ register beats the qubit on both channels (0.53 vs 0.27 on the calibrated ladder, 0.80 vs 0.31 under depolarizing, at $s = 0.005$)—but on the ladder the ordering is no longer monotone in d: the optimum sits at $d = 5$ (0.65), the larger per-event damage of the seventh level pulling $d = 7$ back to 0.53, while under depolarizing $d = 7$ still leads all bases. The intermediate `ion` cost, already split at $d = 5$, now fails cleanly on the ladder (0.07 vs 0.30) and retains only a $+0.07$ edge under depolarizing: the window between native and decomposed cost narrows with every added level, exactly as the rising break-even curve suggests. The criterion itself is conservative here—its $d = 7$ bar of 5.70 is not cleared (the uniform layer ratio is 3.80), yet the $d = 7$ register wins.

V. SCALING WITH PROBLEM SIZE

Figure 4 sweeps precision from 6 to 12.7 bits (control dimensions 64 to 6561; the largest register is the qutrit's fifth size, $m = 8$) at fixed per-layer strength. Both qudits stay above the qubit at every precision on the measured overlap in all four regimes, and the gap over the qutrit widens with size under ladder noise because the qubit decays roughly 3× faster per precision bit: slopes -0.050 ± 0.006/bit (d=2, $R^2 = 0.97$, $n = 4$ sizes) against -0.018 ± 0.008/bit (d=3, $R^2 = 0.66$, $n = 5$) under the calibrated ladder, while the ququint decays at a rate comparable to the qubit's (-0.040 ± 0.004/bit, $R^2 = 0.99$, $n = 3$) from a higher starting point. Two readings deserve emphasis. First, the qutrit's size dependence is the shallowest in every regime, and whether it is flat at all depends on the channel: the fifth size resolves the calibrated-ladder trend to *slow, not flat* (-0.018 ± 0.008/bit, 2.4σ from zero), while under depolarizing the trend remains flat to the largest size (-0.001 ± 0.004/bit, with signal 0.65 ± 0.04 at 12.7 bits). The calibrated-ladder decline is moreover not uniform, and we quote its slope as a summary rather than a model: the first three sizes are flat to $\chi^2/\text{dof} = 0.01$ (0.738–0.742 from 6.3 to 9.5 bits) and the entire fall is carried by the two largest, the 12.7-bit point sitting 3.9σ below the 9.5-bit one. Five sizes cannot separate a constant slope from an onset; the depolarizing family, flat across the same span to 0.5σ, shows no such structure. We report the depolarizing flatness as an observation with a mechanism (Sec. VII C), not a law. One objection must be met first: $D = d^m$ changes across the sweep, so the flatness could be residual grid misalignment drifting with m—the confound of Sec. III returning in a new form. It is not. Residual misalignment is *exactly constant* across the entire sweep for all three bases, on both benchmark instances (0.267 for $d = 2$ and 0.300 for $d = 3, 5$ at $N = 21$; 0.286 for every base at $N = 29$; spread $<10^{-13}$ over the sizes), because $d^m \bmod r$ is periodic in m and the multiset of phase offsets it induces therefore repeats. Alignment does not vary with register size here, so it can explain neither a size-dependent nor a size-independent signal. On the fitted curves the qutrit overtakes the ququint at 7.3 bits (calibrated ladder) and 10.8 bits (depolarizing), and from the smallest common size under the idealized ladder—the story is not monotonic in d, and the best base depends on the target precision. Second, for eigenstate QPE the qudit advantage is decisive and

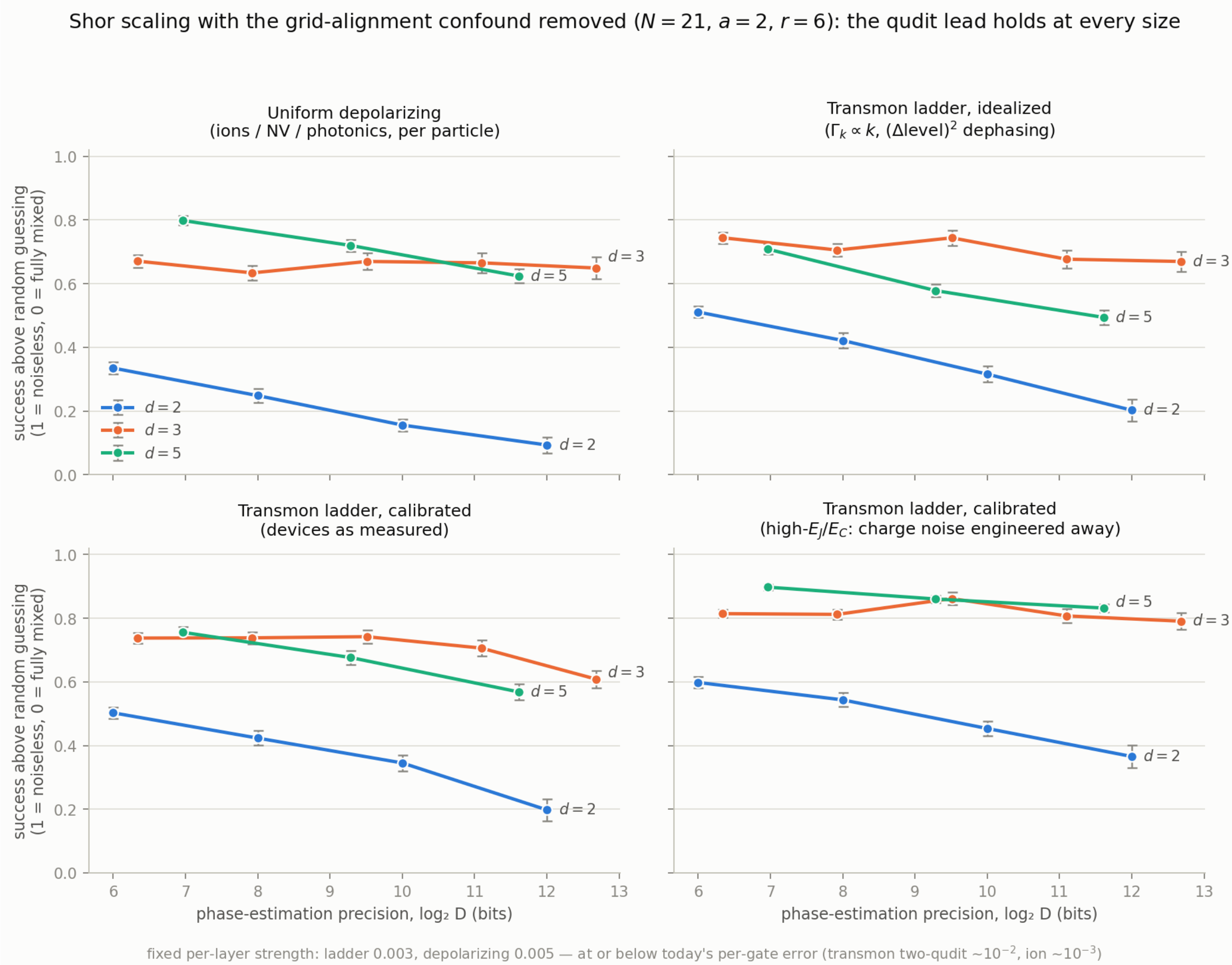

FIG. 4. Register-size scaling on the unbiased instance: floor-corrected signal vs phase-estimation precision (bits), four noise regimes (ladder regimes at strength 0.003, depolarizing at 0.005), `uniform` cost model, registers to 17 qubits, quantum trajectories (1000 per point; 500 at d=2, m=12).

grows with size: at 11.6 bits the ququint retains 2.8× the qubit's signal as measured ($+0.44 \pm 0.02$) and the margin reaches $+0.50 \pm 0.02$ (2.3×) in the high-E_J/E_C regime. Quoted errors are statistical; the dominant uncertainty is systematic—the spread across channel and cost-model choices documented above. Since eigenstate QPE is the quantum-chemistry workhorse, this is the result with the clearest practical consequence.

Instance robustness. Re-running the identical Shor sweep on the second unbiased instance ($N = 29$, $a = 16$, $r = 7$—the exactly alignment-neutral benchmark of Sec. III, whose work register is one digit wider for $d = 3, 5$; 400 trajectories per point) replicates every claim above except one. Both qudits again clear the qubit at every size in all four regimes, and the qubit again decays fastest (-0.040 ± 0.009/bit under the calibrated ladder, matching -0.050 ± 0.006 at $N = 21$; the ququint matches at -0.038 ± 0.001 vs -0.040 ± 0.004). The qutrit's calibrated-ladder slope agrees between the instances within 0.7σ (-0.025 ± 0.004/bit on $N = 29$ against -0.018 ± 0.008 at $N = 21$, five sizes), and is the shallowest in every regime on both. What does not replicate is strict flatness: the $N = 29$ qutrit declines under depolarizing as well (-0.029 ± 0.016/bit), so a flat qutrit family survives only in the depolarizing channel at $N = 21$. What is instance-robust is the ordering, the qubit's fastest decay, and the late qutrit-over-ququint crossing, which recurs near 11 bits on $N = 29$ under the calibrated ladder (7.4 at $N = 21$) and moves beyond the swept range under depolarizing.

VI. GROVER SEARCH: DECOMPOSING THE MECHANISM

Everything above measures one algorithm family—the interpolation experiment of Sec. VII shows Shor *is* phase estimation on an eigenstate superposition. Grover search is the second family, chosen to attack our own weakest claim, with the prediction registered in advance: *qudits win, by less than in phase estimation.* The iteration count $\lfloor(\pi/4)\sqrt{M}\rceil$ is base-independent, so Grover holds the oracle count fixed across bases; its residual depth compression comes only from the narrower multi-qudit decompositions, and at matched size its total exposure compression is roughly half of Shor's (5.7× vs 10.9× from $d = 2$ to $d = 5$). This is a partial rather than a clean separation of width from depth, but it halves the compression while holding algorithm structure fixed. A null result would have meant the advantage was depth all along.

The prediction holds (Fig. 5): the ordering $d = 5 > d = 3 > d = 2$ at every matched size in both noise models, with the Grover advantage at 0.33–0.50 of Shor's. Halving the exposure compression thus costs one-half to two-thirds of the advantage: the compressed register is sufficient for a qudit advantage, and the advantage responds at least proportionally to compression. Grover also serves as an independent methodological control: it has no order, no continued fractions, and an exact $1/M$ floor, so grid alignment is structurally impossible—its agreement rules out the continued-fraction metric as the source of the effect. The cost condition binds more tightly here than for Shor: qudits win under `uniform` and lose under `pavlidis` in both channels, and under `ion` cost at free-evolution ladder dephasing the ququint loses Grover at every matched size and the qutrit above ≈6.5 bits (below which it retains a $\sim 3\sigma$ edge) while the qutrit still wins Shor—the cell where the three algorithms part ways, consistent with Grover's halved compression buying less headroom against the same gate surcharge, and the cell that refocusing repairs (Sec. VIII).

VII. THE MECHANISM: DAMAGE UNITS AND THE DECODER

A. Entanglement: a falsified hypothesis

Shor entangles control and work registers; eigenstate QPE does not. An early hypothesis attributed the (confounded) Shor/QPE difference to which-path dephasing through the work register. Because Shor's work input $|x{=}1\rangle$ is an equal superposition of the r eigenstates of the modular multiplier, QPE on a K-fold eigenstate superposition interpolates continuously from eigenstate QPE ($K = 1$) to the Shor regime ($K = r$) with circuit, metric, noise, and cost held fixed; the control–work entanglement entropy is exactly $\log_2 K$ bits. The measured effect is real but an order of magnitude too small: ≈ -0.025 signal per bit, in all four noise/cost conditions, against the ≈0.5 gap—measured on the then-current, confounded $N = 15$ comparison—it was meant to explain (that gap itself did not survive the de-confounding of Sec. III). We report it as a falsified mechanism—and it was chasing this failure that exposed the grid-alignment confound of Sec. III.

TABLE II. The collapse test: goodness of a single two-parameter exponential Ae^{-kx} pooled over both algorithms, all bases and sizes.

ordinate	abscissa	ladder R^2	depol. R^2
signal	exposure × strength	0.67	0.79
signal	exposure × damage	0.81	0.79
fidelity	exposure × damage	**0.97**	**0.99**

B. Exposure in damage units, and the decoder boundary

The surviving mechanism candidate is total noise exposure: carriers × time-layers. If it were sufficient, signal would be a function of exposure × strength alone and all points—both algorithms, all bases, all sizes—would collapse onto one curve. They do not ($R^2 = 0.67/0.79$ for ladder/depolarizing), and the failure decomposes into two identifiable pieces.

Piece 1: exposure must be counted in damage, not events. One noise layer does d-dependent harm. Writing S for the one-layer superoperator, the per-carrier-layer entanglement infidelity is $1 - F_{\mathrm{e}}$ with $F_{\mathrm{e}} = \operatorname{tr} S/d^2$: under the calibrated ladder at strength s it is $0.75s$, $1.46s$, and $2.82s$ for $d = 2, 3, 5$—a ququint takes $\sim 4\times$ a qubit's damage per event—and $p(1 - 1/d^2)$ under depolarizing. With the abscissa exposure × $(1 - F_{\mathrm{e}})$, Grover's three bases collapse onto a single exponential: per-family decay rates 0.44/0.49/0.43 (a 1.1× spread, against 3.6× in event units), each family log-linear with $R^2 \geq 0.996$. The reweighting is a ladder effect: depolarizing damage is already nearly flat in d ($0.75p$–$0.96p$), and there the signal rows of Table II gain nothing from it.

Piece 2: the residual is the decoder, not the noise. With units fixed, a nested fit finds one shared amplitude with a per-*algorithm* decay rate at $R^2 = 0.93/0.94$ (ladder/depolarizing). The split is stark at the family level: Grover is a textbook exponential in damage, while Shor's $d = 3$ family holds a flat signal of ≈0.73 while its exposure triples. A candidate explanation is that Grover's "decoder" is the identity—its signal *is* marked-state survival—whereas Shor's signal passes through continued-fraction order recovery, whose error tolerance grows with register size: added digits add exposure but also add redundancy the decoder exploits.

This is directly testable: if the decoder story is right, end-state *fidelity* $\langle\psi_{\mathrm{ideal}}|\rho|\psi_{\mathrm{ideal}}\rangle$ should obey the one-curve law that signal does not. It does (Table II): one shared exponential in damage units fits the fidelity of

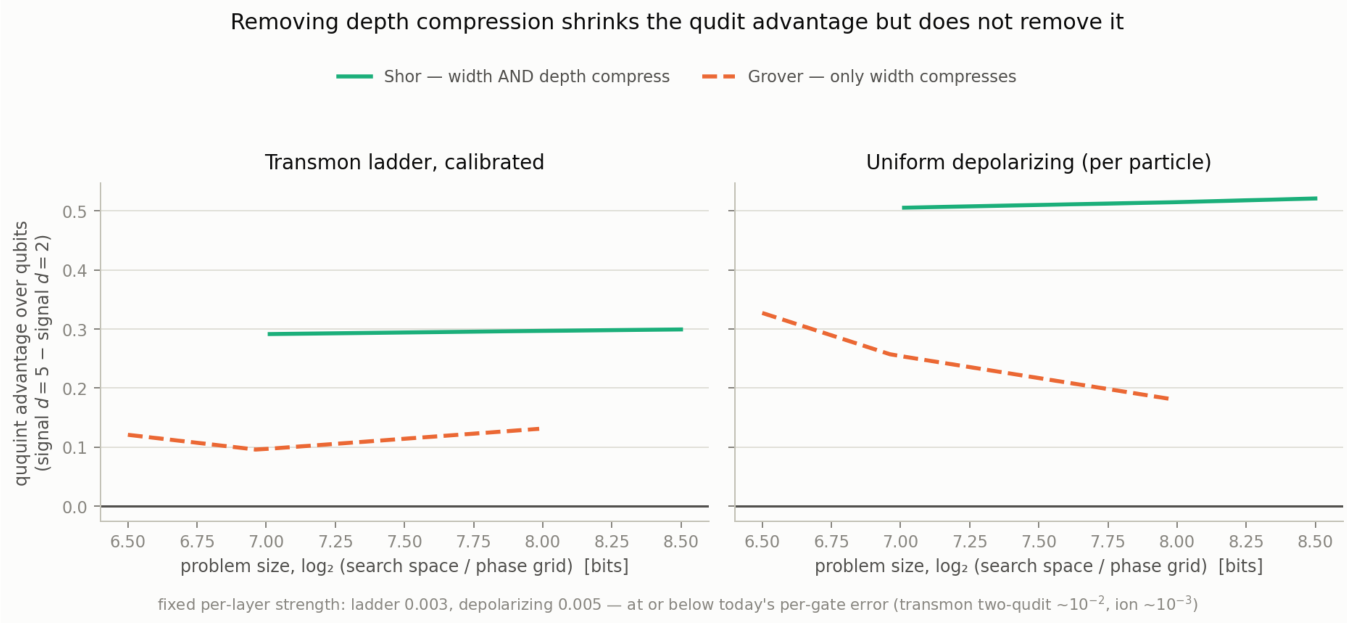

FIG. 5. Grover vs Shor floor-corrected signal at matched problem size, both noise models, `uniform` cost model (native entangler). The qudit ordering $d{=}5 > d{=}3 > d{=}2$ holds at every size in both models; the Grover advantage is 0.33–0.50 of Shor's, consistent with its halved exposure compression.

both algorithms at $R^2 = 0.97$ (ladder) and 0.99 (depolarizing), with fitted amplitude ≈ 1 as the zero-noise limit demands; granting the algorithms separate rates adds only 0.02/0.003. The decoder gap is visible in the raw numbers: Grover's fidelity equals its signal to better than 0.006 under the ladder and 0.025 under depolarizing at every point, while Shor's $d = 3$ family loses two thirds of its fidelity ($0.55 \to 0.18$) with signal flat, and at $d = 2$, $m = 12$ under depolarizing, continued fractions decode a signal of 0.093 ± 0.025 from a state with fidelity 3×10^{-4}. The mechanism claim, in the form the data support: *accumulated channel damage is the law for the quantum state; the residual algorithm dependence of decoded success sits in the decoder*—for which Sec. VII C gives an exact quantitative account, derived and verified against the decoder itself.

What the collapse does and does not prove. For a product of near-identity incoherent channels, log-fidelity is additive in the per-application entanglement infidelity to first order, so a single exponential with amplitude ≈ 1 is the *null expectation*, not a discovery. The informative content of Table II is therefore bounded as well as measured. In linear fidelity the shared fit gives $R^2 = 0.97/0.99$; rescored in *log* fidelity—the metric that weights the deep tail—the same fit gives $R^2 = 0.76$ in both channels (0.85–0.91 if refit there). The law holds within a factor of two down to fidelity 1.5×10^{-2} on the ladder and, excepting a single floor-pinned Grover point, to 2.7×10^{-4} under depolarizing; every departure is a Grover point approaching its $1/\dim \mathcal{H}$ fidelity floor (the deepest, at 5.3×10^{-3}, sits $1.4\times$ above its floor of 1/256 and $82\times$ above the fit)—a floor that varies across exactly the bases and sizes the collapse unifies. What survives everywhere is the decoder residual, which no channel-level argument predicts. Conversely, the law's failure mode is error that does not compose incoherently: correlated or coherent noise breaks the additivity, and that is precisely what the deep hardware circuit exhibits (Sec. IX)—which is why the law is stated for Markovian incoherent channels only (see Limitations).

C. Why the decoder gains tolerance with size

The size dependence of Shor's decoder tolerance has a closed form, and because the decoder is classical we can measure it exactly rather than argue it. Continued-fraction recovery succeeds for any outcome y whose phase y/D lies within $1/(2r^2)$ of some s/r—the classical convergent guarantee [22, 33]. This sufficient condition gives an acceptance window around each of the $r - 1$ peaks spanning $1/r^2$ in phase and holding $\sim D/r^2$ outcomes, while the noiseless interference peak stays ~ 1 outcome wide at every size. The window therefore grows linearly in $D = d^m$, exponentially in register size, whereas noise-induced broadening grows only polynomially in m (exposure = carriers × layers). The decoder's redundancy outruns the added exposure, which is why Shor's signal families sit flat while their fidelity decays exponentially—and why Grover, whose acceptance set is the single marked item at every size, shows no such flattening.

Table III runs the project's own continued-fraction routine over every outcome $y \in [0, D)$ and counts the acceptance set exactly; the growth from 1.6 accepted out-

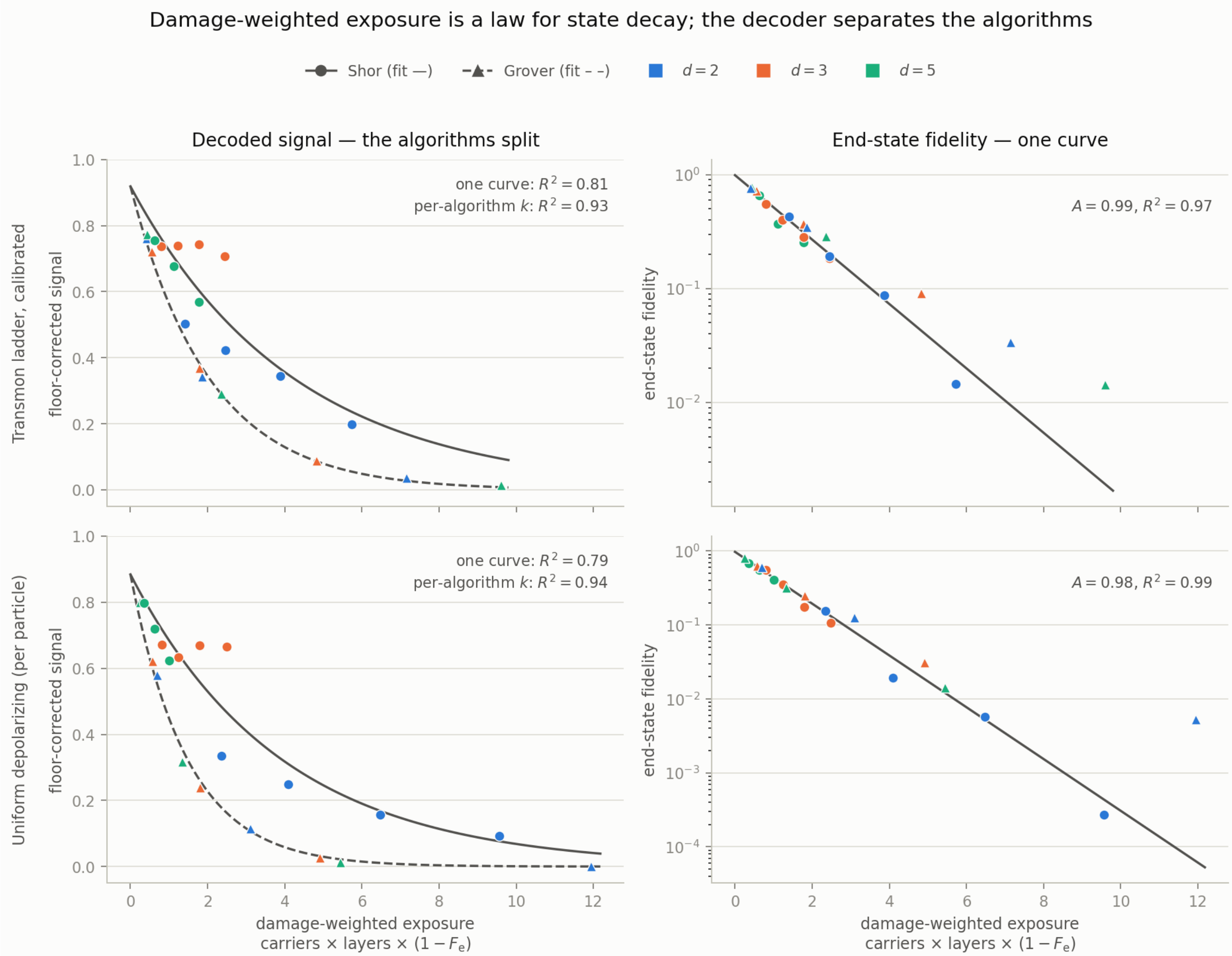

FIG. 6. Damage units and the decoder. Left column: floor-corrected decoded signal against damage-weighted exposure—the algorithms refuse a common curve ($R^2 = 0.79$–0.81), and granting a per-algorithm decay rate (solid: Shor; dashed: Grover) recovers $R^2 = 0.93$–0.94; Shor's $d = 3$ family (orange circles) sits nearly flat while its exposure triples. Right column: end-state fidelity on the same abscissa (log scale) falls on a single exponential with amplitude ≈1 for both algorithms, all bases and sizes ($R^2 = 0.97/0.99$; `uniform` cost model). The residual algorithm structure in decoded signal is therefore the classical decoder, not unexplained noise physics.

comes at $D = 64$ to 116 at $D = 4096$ is a 73× gain in tolerance against a peak that never widens. The acceptance set moreover has an exact description. The decoder accepts y if and only if some convergent denominator q of y/D is a multiple of r with $q \leq N$ (Appendix A proves the equivalence and reconciles it with the divisor-form recovery of the standard analyses), which reduces A to pure number theory: for a reduced fraction p/q, the phases with p/q among their convergents form the open interval between the Stern–Brocot mediants $(p+p')/(q+q')$ and $(2p-p')/(2q-q')$, where p'/q' is the penultimate convergent of p/q—a classical fact of continued-fraction theory [33, 34]. Partitioning outcomes by the *first* admissible convergent of y/D makes the decomposition disjoint (the side denominator $q - q'$ is never a multiple of r, since $\gcd(q', q) = 1$), and summing the interval counts reproduces the enumerated $|A|$ *without error*—outcome for outcome, on all 27 instance/size combinations tested, including all four within-modulus orders. Summing interval measures instead of counts gives the asymptotic law

$$\frac{|A|}{D} \longrightarrow 2\ln 2 \sum_{k=1}^{\lfloor N/r \rfloor} \frac{\varphi(kr)}{(kr)^2}, \qquad (2)$$

with φ Euler's totient (the exact per-denominator measure is $\mu(q) = \frac{2}{q}\sum_{u<q,\,\gcd(u,q)=1}(q+u)^{-1} \to 2\ln 2\,\varphi(q)/q^2$; summed over *all* denominators $q \leq Q$ this weight reproduces the almost-everywhere count $(12\ln 2/\pi^2)\ln Q$ of convergent denominators below

TABLE III. The decoder measured, not estimated: exact size of the continued-fraction acceptance set $A = \{y \in [0, D) : \mathrm{decode}(y) = r\}$ on the benchmark instance ($N = 21$, $r = 6$), obtained by running the decoder over every outcome. "per peak" is $|A|/(r-1)$; the envelope estimate is D/r^2; "law" is the totient sum of Eq. (2) per peak, whose exact finite-D form (see text) reproduces $|A|$ without error at every size. No simulation is involved.

d	m	D	$\lvert A\rvert$	per peak	D/r^2	law
2	6	64	8	1.6	1.8	1.8
2	8	256	36	7.2	7.1	7.2
2	10	1024	148	29.6	28.4	28.9
2	12	4096	582	116.4	113.8	115.7
3	4	81	10	2.0	2.2	2.3
3	5	243	36	7.2	6.8	6.9
3	6	729	100	20.0	20.2	20.6
3	7	2187	308	61.6	60.8	61.8
5	3	125	16	3.2	3.5	3.5
5	4	625	88	17.6	17.4	17.6
5	5	3125	442	88.4	86.8	88.2

Q [34], so the measure itself is classical—the content of Eq. (2) is its restriction to the denominators the decoder admits); it holds to better than 1% on five of our six instances at $D \gg N^2$, and to 4.2% on the sixth ($N = 55$, $r = 5$, where eleven admissible denominators make the first-admissible exclusions largest). The law settles both scaling directions at once. In D it is exactly linear—the measured $|A| \propto D^{1.03\pm0.01}$ ($R^2 = 0.999$) carries its small excess over slope 1 purely from window discreteness. In r it replaces the $1/r^2$ envelope: comparing $r = 5$ with $r = 10$ on identical registers (the within-modulus pairs of Sec. III, swept to $D \geq 5r^2$ so the window exceeds one outcome) gives measured per-peak ratios of 9.6 ($N = 33$) and 8.9 ($N = 55$) where $1/r^2$ predicts 4.0; the law's totient-sum ratios are 9.7 and 9.3. It also explains the envelope's apparent success in Table III as an accident of the benchmark instance: at $r = 6$, $N = 21$ the naive $(r-1)/r^2 = 0.139$ sits within 2% of the law's 0.141 because two errors compensate—the true $q = r$ window measure $2\ln 2\,\varphi(r)/r^2$ is only $0.55\times$ the envelope's, and the admissible multiples $2r, \ldots, \lfloor N/r\rfloor r$ restore the difference. On the $r = 5$ instances, where the cancellation fails, the envelope is off by 2.5–3× while the law stays within 4.2%.

Relation to success-probability analyses of order finding. An exact literature bounds the success of continued-fraction post-processing on base-2 registers: Shor's original $4/\pi^2$ asymptotics [22], the sharp divisor-recovery bounds of Gerjuoy [35] and of Bourdon and Williams [36], Ekerå's proof that a single run recovers r with probability approaching one [37], the tight two-sided window bounds of Magdon-Ismail and Dong [38], the self-contained treatise of Barzen and Leymann [39], and, for eigenstate phase estimation, Chappell *et al.* [40]. These analyses score outcomes inside a tolerance window of the peaks via the convergent guarantee—a sufficient condition—and typically certify recovery of a divisor $r/\gcd(s, r)$ that a classical search then lifts to r. The law above differs in kind: it is an exact, outcome-for-outcome description of the full acceptance set of a decoder that must return r itself, including the contribution of the admissible denominators $2r, \ldots, \lfloor N/r\rfloor r$ that window bounds do not count. Eq. (2) depends only on r and N, not on the base: at matched control dimension the decoder's tolerance is *identical* across bases, so the entire cross-base difference in decoded success sits in the quantum state—precisely this section's mechanism claim. We are not aware of this acceptance set having been computed before for a decoder required to return r itself.

This decomposition has a general reading beyond the present comparison. A decoded success probability conflates two quantities with different physics: the decay of the quantum state, which damage-weighted exposure describes universally, and the error tolerance of the classical post-processing wrapped around the measurement, which is algorithm-specific and can grow with problem size. Benchmarks of "noise resilience" across algorithms or encodings that use decoded success as their metric are therefore partially measuring decoder redundancy—a classical resource. Reporting end-state fidelity alongside decoded signal, as in Table II, separates the two at no extra simulation cost, and we suggest it as standard practice for noisy-algorithm comparisons.

VIII. ROBUSTNESS

Readout error. A d-level readout must resolve d pointer states, the higher ones worst. Charging a readout channel with misread rate of $|k\rangle$ growing as $(1 + k)$—a linear reading of the few-percent, higher-is-worse readout errors reported for transmon qutrits [2, 3]—on every control carrier of every base leaves the qudit advantage untouched: the ququint's lead drifts by less than ± 0.03 over the full sweep $\varepsilon = 0$–0.04 (an 8× range among nonzero settings). The reason is a structural cancellation: total readout exposure is $m \times \varepsilon(d{+}1)/2$, and at matched precision $m \approx \log D/\log d$, giving $9\varepsilon/8\varepsilon/9\varepsilon$ for $d = 2/3/5$ at $D \approx 64$. Per-level readout degradation with d is almost exactly cancelled by the reduced carrier count—near-neutral by construction, not luck. The neutrality claim is scoped to simultaneous d-level discrimination (the transmon case that motivates the $(1{+}k)$ model): ion-qudit readout is sequential shelving with $d-1$ detection rounds, so misreads compound down the sequence and readout *time* grows with d, adding idle decoherence the cancellation above does not include.

Refocusing. No transmon runs a long circuit without echo, so the honest question is whom it helps more. Sweeping the dephasing scale from free evolution to perfect echo (the same knob that models the high-E_J/E_C regime [27]), the ququint's lead grows monotonically in all four conditions tested. The consequential case is linear-in-d gate cost: the ququint *loses* Shor without echo (-0.026) and *wins* with it ($+0.191$); QPE moves from exact parity to $+0.196$. Mechanistically, dynamical decoupling suppresses dephasing—the part of the ladder channel scaling worst with d (the max-level law)—and leaves the gentler $k^{0.7}$ relaxation, in agreement with qudit DD experiments [26] and with the refocused molecular-qudit QFT of Ref. [41]. This is why the central condition is stated "at the device's operating dephasing level": refocusing buys roughly one cost model of headroom, and benchmarks of qudit algorithms should be run on refocused devices.

Noise-inflation threshold. The channels charge every base the same per-layer strength; if higher-d gates are additionally noisier per layer, the condition acquires a threshold. Inflating the qudit's per-layer strength alone, $s_d = f\, s_2$, and locating the crossing f^* at which the qudit's floor-corrected signal falls to the qubit's (demo instance, exact density matrices, both operating points): under native-gate cost $f^* = 2.0$–2.5 on the calibrated ladder and 2.6 (d=3) to 4.5 (d=5) under depolarizing; under the linear `ion` cost $f^* = 1.2$ (d=3, ladder) and 1.6 (d=3, 5, depolarizing), the ladder ququint cell being already lost at $f = 1$ (Sec. IV). Read as hardware guidance: a platform keeps the qudit advantage while its measured qudit-to-qubit per-gate noise ratio stays below f^* *after* the layer-count multiplier is charged—roughly 1.6× for today's trapped-ion pairing and 2.0–2.5× for a native-entangler transmon (the calibrated-ladder/`uniform` pairing; the 4.5× cell belongs to depolarizing/`uniform`, a crossed pairing).

Zeeman-structured dephasing. The per-particle convention flattens pair structure the ion encoding really has, so we also test its worst case: magnetic-field dephasing carrying the collective-B sensitivity structure of the $^{40}Ca^+$ level indexing of Ref. [1], applied independently per carrier—the uncorrelated worst case—realized exactly by a single diagonal jump $\propto \mathrm{diag}(g_j m_j)$ ($g_S = 2$, $g_D = 6/5$; the Markovian generator stands in for the unrefocused effect of what is, in the laboratory, quasi-static noise), normalized so the optical-qubit pair $0\leftrightarrow 1$ matches the qubit channel. Pair rates then span 1–25× that reference at $d = 3$ and 1–49× at $d = 5$. The encoding is no strawman: over all $\binom{8}{d}$ level subsets of the $S_{1/2}/D_{5/2}$ manifold, the chosen levels exactly minimize the worst pair rate at $d = 5$ and $d = 7$; only $d = 3$ admits a gentler choice (9×, via $\{S_{-1/2}, D_{-3/2}, D_{-1/2}\}$). The verdict reverses outright: the qubit wins at every strength under both cost models (0.72 against 0.24/0.23 at $s = 0.003$, `uniform`). Nor is the missing correlation structure doing the work: repeating the study with genuinely common-mode dephasing—one global jump coupling to the total Zeeman shift, exact as an elementwise mask on the density matrix—preserves the reversal (0.87 against 0.45/0.34 at the same point; its decoherence-free pairs soften every base, the qubit most). Unmitigated Zeeman-structured dephasing on a sensitivity-ordered encoding is therefore a regime with no qudit advantage at all—the sharpest failure mode we found, and one the condition's "operating dephasing level" clause must be read to include. Measured devices are engineered away from this regime: shielding makes coherence times of order 100 ms achievable across all transitions [1]. The device's randomized benchmarking—per-pulse error rising only 1.6× from $d = 3$ to $d = 5$ (2.0×10^{-4} to 3.2×10^{-4}) where the raw sensitivity ratio is 25–49—is consistent with a near-flat residual, though a few-microsecond pulse cannot resolve slow dephasing; the direct evidence the convention needs is d-resolved Ramsey or echo coherence across the encoding's transitions, and pending that, the per-particle convention is a statement about mitigated hardware whose price of failure the following sweep fixes. The mechanism mirrors the refocusing result above—structured, slow dephasing is exactly the part echoes remove—and the price is quantitative: composing the Zeeman component with the depolarizing operating point ($s = 0.005$; the two channels commute, so the product is exact) and sweeping the suppression factor ε applied to the unmitigated Zeeman reference ($s_Z = 0.003$ per carrier-layer), the qudit ordering returns at $\varepsilon^* = 0.58$–0.79 under native-gate cost but only at $\varepsilon^* = 0.09$–0.15 under the $2(d-1)$ Mølmer–Sørensen cost. In hardware units the bar is a $0\leftrightarrow 1$-pair coherence of $\gtrsim$400–600 layer times for native-gate cost and $\gtrsim$2200–3800 for MS cost: at $\sim$100 μs layers, a 100 ms shielded coherence meets the native-gate bar with a factor-two margin but falls short of the MS-cost bar by 2–4×. On linear-cost hardware the Zeeman component must be suppressed below the plain shielding level—deeper echo sequences or magnetically insensitive encodings—before the qudit ordering returns.

Composite dimension. Nothing in the circuits or channels uses primality. A $d = 4$ point on the demo instance lands in the qudit band (0.83/0.75 against 0.73/0.67 at $d = 3$ and 0.76/0.78 at $d = 5$), but $d = 4$ is a prime power—GF(4) exists, so the fault-tolerant machinery does not break there—and its $D = 64$ register shares the qubit's residual misalignment bit for bit, so it cannot settle the question alone. The decisive control is $d = 6$—composite, non-prime-power, no field structure—run on the alignment-neutral instance ($N = 29$, $r = 7$; residual misalignment 0.2857, identical for every base $d = 2$–6; 1000 trajectories per point): the family reads 0.45/0.64/0.77/0.70/0.77 ($d = 2/3/4/5/6$, calibrated ladder) and 0.29/0.53/0.72/0.71/0.84 (depolarizing), both composites inside the qudit band and $d = 6$ at its top under depolarizing. The bare-circuit dynamics carries no trace of primality; the prime restriction elsewhere in this paper is inherited from the fault-tolerance motivation [7, 9].

Independent validation. Beyond internal checks (Sec. XI), we reproduce the three central equations of Ref. [30]—qudit and multi-qubit process infidelities and the critical curve $(d^2-1)/(3 \log_2 d)$—from our superoperator code with no analytics of our own, to a worst relative error of 4.1×10^{-4} over $d = 2$–64, with the residual identi-

TABLE IV. Hardware runs vs compiled-circuit predictions. "predicted" is the exact-DM band at the vendor-reported two-qubit infidelity ($s = 0.007$ for Forte-1), spanning the two exposure conventions; Garnet predictions are not quoted because server-side SWAP routing makes the compiled depth compiler-dependent. Forte-1 main runs use IonQ debiasing over 5,000 shots; success is scored in the digit-reversed reading (see text). $|w\rangle$ is the work-qubit $|1\rangle$ population.

device	circuit	predicted	measured	$\lvert w\rangle$
Forte-1	m=5 (15 gates)	0.60–0.70	0.617 ± 0.007	0.99
Forte-1	m=7 (28 gates)	0.42–0.54	0.011 ± 0.001	0.99
Forte-1	m=7 AQFT (25)	0.44–0.57	0.066 ± 0.005	0.99
Garnet	m=5 (15 gates)	—	0.080 ± 0.004	0.81
Garnet	m=7 (28 gates)	—	0.032 ± 0.003	0.66

fied as their first-order truncation (it tracks the infidelity itself).

IX. HARDWARE ANCHOR

The simulation stack makes falsifiable predictions, and the qubit rows of Table V compile at face-value gate counts on cloud hardware: with the work register reduced to one qubit prepared in the eigenstate, each controlled-U^{2^i} is a single controlled-phase, and the no-swap inverse QFT adds $m(m-1)/2$ more—15 entangling gates at $m = 5$ (6 qubits), 28 at $m = 7$ (8 qubits). We ran these circuits on two commercial devices (Table IV): the 36-qubit trapped-ion IonQ Forte-1 (all-to-all connectivity, vendor-reported two-qubit ZZ fidelity 0.993) and the 20-qubit square-lattice superconducting IQM Garnet. Predictions were recomputed for the compiled circuits by exact density-matrix evolution under the paper's per-particle depolarizing convention, bracketed by two exposure accountings (every gate a full noise layer, or single-qubit gates charged their measured 130/970 duration ratio).

Three findings. *First, the shallow ion circuit lands inside the predicted band*: 0.617 ± 0.007 against 0.60–0.70. Debiasing is an error-mitigation ensemble average over qubit mappings and is not doing the work here: the raw 1,000-shot pilot scored 0.608 ± 0.015, within 0.009 of the debiased value and inside the same band, so the bare-channel prediction is met by an essentially unmitigated number. Because the success-vs-strength map is steep, this single number pins the device's effective per-gate depolarizing strength at 0.007–0.009 (depending on exposure convention), bracketing the vendor's measured 0.7% two-qubit infidelity—a quantitative validation of the noise convention used throughout this paper, on the hardware class it models. *Second, the deep ion circuit fails coherently, not by decoherence.* At $m = 7$ the interference peak is destroyed (0.011, below the random floor of 0.031) while the work qubit remains at 0.99; no relabeling of the outcome bits recovers the peak (the best of 10,080 reinterpretations reaches only 0.21, not significant over so large a hypothesis set), and a 500-shot raw $m = 4$ probe caught the mechanism directly, unmitigated—a nearly pure output state (71% of shots on one outcome) whose phase is wrong by one least-significant digit. Debiasing does not remove the failure the raw probe establishes, and an approximate-QFT variant that drops the three smallest rotation angles recovers only a factor $\sim$6 (0.066 vs a 0.44–0.57 band): the deep no-swap QFT fails wholesale on this device, consistent with accumulated coherent angle systematics rather than the incoherent noise our channels describe. (A base-2 datum adjacent to the truncation-robustness conjecture for qudits [14]—which it cannot test directly: truncation buys an order of magnitude but cannot repair a coherent failure.) *Third, the superconducting lattice fails by plain decoherence*: both circuits sit at the random floor with the work qubit visibly decayed (0.81/0.66), the signature of SWAP-routing the all-to-all kickback pattern through a fixed-connectivity chip whose coherence time is four to five orders shorter than the ions'.

The three-way outcome maps onto the paper's framework. The per-particle depolarizing convention quantitatively describes the platform it was calibrated to—provided the circuit is shallow enough that coherent systematics stay subdominant, the regime our Limitations already fence off. The superconducting failure is compilation overhead deciding the outcome before noise structure enters, the hardware face of the `pavlidis` lesson. And the deep-circuit ion failure is a caution for any decoded-success benchmark on NISQ hardware: past a depth threshold, what is measured is coherent calibration error, not the decoherence such benchmarks nominally probe. One practical note for reproduction: IonQ returns the control register digit-reversed relative to the Braket local-simulator convention (the $m = 5$ peak sits exactly on the bit-reversed ideal outcome); all scores above use that pre-registered reading.

X. DISCUSSION

Which platform first. Both physically matched cost/noise pairings favor ququints under their mitigated operating conventions, but their robustness to residual structured dephasing differs sharply (Sec. VIII): the native-entangler route tolerates a Zeeman component up to $\varepsilon \approx$ 0.6–0.8 of the unmitigated reference, the linear-MS-cost route only 0.09–0.15. The transmon route is therefore the robust one; the ion route offers the larger headline gains on eigenstate QPE, conditional on collective-B suppression beyond plain shielding—a requirement that belongs to the experimental specification below. The cheapest decisive experiment we can propose is eigenstate QPE at $d = 5$, $m =$ 2–3 on a Ringbauer-class ion processor [1]: six native entangling gates at $m = 3$ ($\sim$8–12 two-qudit gates after reduction,

TABLE V. Predictions for the proposed trapped-ion experiment: eigenstate QPE under per-particle depolarizing noise, `ion` cost model, exact density-matrix evolution (no sampling error). Strengths span the projected single-qudit figure ($s = 10^{-3}$) and the band of *demonstrated* two-qudit entangling gates (5×10^{-3}–2×10^{-2}: two-qutrit Mølmer–Sørensen fidelities of 95–98%, none published above $d = 3$). Pairs are matched in control dimension D and scored as the phase correct to b bits ($b = 4$ is the finest criterion the $D = 25$ grid can express). "gates" counts native entangling gates, with total entangling time-layers under the $2(d-1)$ Mølmer–Sørensen cost in parentheses. s is the per-carrier-layer strength of Sec. II: a two-qudit gate spans $d - 1$ layers, so its deposited noise is $\sim 2(d-1)s$ on the gated pair—read against measured gate fidelities, the bands are per-pulse-equivalent, the conservative direction for the qudit rows. Readout is not charged; sequential-shelving readout adds up to $d-1$ detection rounds at $\sim 3 \times 10^{-3}$ worst-case qudit error [1], a d-dependent cost the experimental budget must carry.

register	b	gates	noiseless	s=0.001	0.005	0.01	0.02
d=5, m=2	4	3 (20)	0.816	0.785	0.674	0.562	0.401
d=2, m=5	4	15 (30)	0.917	0.807	0.500	0.299	0.142
d=5, m=3	5	6 (36)	0.948	0.855	0.575	0.364	0.168
d=2, m=7	5	28 (49)	0.989	0.778	0.328	0.140	0.053

i.e. 64–96 Mølmer–Sørensen pulses at $2(d-1)$ per gate—demanding at demonstrated per-pulse fidelities, but landing in the s band of Table V where the predicted separation is largest), on a platform where no multi-qudit algorithm requiring entangling gates has yet been run above $d = 3$ [42, 43]—single-qudit demonstrations reach $d = 8$ on a trapped ion [43] and, on photonic platforms, $d = 32$ Shor and qudit QPE [44, 45]. Table V supplies the predicted success probabilities, computed by exact density-matrix evolution with no sampling error. At the measured ion operating point ($s = 10^{-3}$) the $m = 3$ ququint register is predicted to succeed at 0.855 against the matched qubit register's 0.778, a raw gap that doubles at twice the operating strength. The $m = 2$ pair is subtler: the coarse $D = 25$ grid depresses the ququint's noiseless ceiling (0.816 vs 0.917), so its advantage appears in floor-corrected signal (0.96 vs 0.87 at the operating point) while raw success crosses only near $s = 2 \times 10^{-3}$—the $m = 3$ pair is the decisive one, and either outcome falsifies a side of the condition. Crucially, the proposal does not wait on projected gate fidelities: demonstrated two-qudit entangling gates sit an order of magnitude above the single-qudit figure, and in that band the separation *widens*—0.575 vs 0.328 at $s = 5 \times 10^{-3}$ and 0.364 vs 0.140 at 10^{-2} for the $m = 3$ pair (36 against 49 entangling layers; six native gates against the qubit's twenty-eight)—while the $m = 2$ raw-success crossing at 2×10^{-3} means today's noisier gates already put the ququint ahead in raw success as well (0.674 vs 0.500 at 5×10^{-3}). The deep-circuit coherent failure of Sec. IX counsels entering at the shallow $m = 2$ pair before the decisive $m = 3$ one.

Relation to code-level comparisons. Keppens *et al.* [28] find slightly *worse* qudit logical error rates for the five-qudit code under the same noise convention we use, and the two results are compatible: a code is five carriers and a fixed gate list at every q—no problem instance to compress—so raising q buys no width or depth while charging every carrier more. An algorithm compresses; that compression is the entire source of the advantage measured here. A qudit advantage must be argued at the level of a compressible workload, and does not transfer automatically to the error-correction layer. Relatedly, Bocharov *et al.* [17] show emulated-binary arithmetic can beat native ternary—encoding density alone is not a mechanism—which is consistent with our attribution of the advantage to width-and-depth compression under a per-carrier noise budget, and with its disappearance when compilation buys the compression back.

Recommendations to the field. Cross-dimension algorithm comparisons should (i) randomize or report grid alignment (an irrational target phase does it for QPE; for order finding, choose r dividing no d^m and report residual misalignment); (ii) match problem size across bases by interpolation rather than comparing raw d^n points; (iii) charge multi-qudit gates their decomposition depth; and (iv) report the floor-to-baseline span of the metric (we require >0.15; Sec. II discloses our one grandfathered exception), since small orders collapse the continued-fraction metric's dynamic range. Each rule was learned by stepping on the corresponding rake.

Limitations. Our channels are Markovian and incoherent: coherent errors (cross-Kerr shifts $\alpha \approx 0.1$–$0.7\,$MHz, drive-induced shifts) and leakage during gates are not modeled; correlated noise enters only through the common-mode Zeeman test of Sec. VIII. Cross-Kerr in particular is known to be fast on unprotected qutrits [3], and since the number of level pairs it can act on grows with d, its omission plausibly favors qudits—the sign of this bias runs against our conclusion, not for it. Gate cost and gate error are treated as proportional (more layers = more idle decoherence); a model where longer gates achieve better per-operation fidelity would soften the ion penalty, and the inflation threshold of Sec. VIII bounds the tolerance in the opposite direction. The schedule is serial (every carrier idles during every gate), matching single-addressed ion strings; platforms that execute gates concurrently would cut idle exposure most for the widest register, i.e. for $d = 2$, softening the qudit advantage there. The scaling trends are measured to 17 qubits and stated as slopes with error bars, not theorems: the Shor sweep uses two instances ($N = 21$, $r = 6$ and $N = 29$, $r = 7$) at one noise strength per regime, with 3–5 sizes per base. The ordering and the qubit's fastest decay replicate across the instances, as does the qutrit's calibrated-ladder slope within 0.7σ (-0.018 vs -0.025/bit); individual per-base slopes nonetheless carry instance-level variance of order 0.01/bit. A fifth qutrit size ($m = 8$, 12.7 bits) resolves the $N = 21$ calibrated-ladder trend to slow rather than flat, and locates the fall in the two largest sizes rather than spreading it across the sweep—

a shape five sizes can report but not fit; strict flatness persists only under depolarizing. The qutrit's decoder account is derived and verified exactly (Sec. VII C) and grid alignment is excluded as a confound. Statistical error bars are small throughout; the dominant uncertainty is the systematic spread across channel and cost-model choices, which is why we report the full grid. Dimensions beyond $d = 7$ are untested, and $d = 7$ itself only at demo size on one instance (Sec. IV), where the break-even window has visibly narrowed. All simulations are of ideal algorithm circuits: the controlled-U^{d^i} multipliers are applied as exact unitaries, each charged one gate at the cost model's per-gate rate, whereas compiled modular arithmetic—depth $4d^2q$ per adder under two-level decomposition [14]—has a d-dependence of its own: at face value the compiled qudit-to-qubit depth ratio spans $\approx$0.9–2.7 across $d = 3, 5$ (depending on whether adder- or full-exponentiation depths dominate), extending beyond the `pavlidis` row's 1.0–1.65, so compiled arithmetic can be harsher than any cost model charged here and the decomposition verdict is, if anything, conservative. Fault-tolerant overhead is out of scope, and per Ref. [28] its dimension-dependence may differ in sign. Finally, $d = 5$ entangling gates exist today only on trapped ions—our deep-register results are one to two hardware generations ahead of experiment.

XI. METHODS

Simulators. Exact density-matrix evolution (states as rank-$2n$ tensors, channels applied per carrier per layer) to Hilbert dimension $\sim$3000; beyond that, Monte Carlo wavefunction trajectories [46, 47]: after each gate, each carrier independently passes through the per-layer channel raised to the gate's cost, sampled via one Kraus operator drawn with probability $\mathrm{tr}(K^\dagger K \rho_q)$ from the carrier's reduced state. Averaging $|\psi\rangle\langle\psi|$ over trajectories reproduces the channel exactly; each trajectory contributes its full outcome distribution, so statistical error is well below Bernoulli. Fractional gate costs are exact: both channel families form one-parameter semigroups, so E^t is evaluated by scaling the Lindblad rate (ladder) or as $1 - q = (1 - p)^t$ (depolarizing). End-state fidelity is estimated as the trajectory average of $|\langle\psi_{\mathrm{ideal}}|\psi\rangle|^2$, an unbiased estimator of $\langle\psi_{\mathrm{ideal}}|\rho|\psi_{\mathrm{ideal}}\rangle$, cross-checked against exact density-matrix evolution. Throughout, $F_{\mathrm{e}} = \mathrm{tr}\, S/d^2$ with S the one-layer superoperator in the natural representation; for a channel with Kraus operators $\{K_i\}$ this is $\sum_i |\mathrm{tr}\, K_i|^2/d^2$, the entanglement fidelity of the channel with respect to the maximally entangled state.

Calibrated dephasing via Euclidean embedding. With diagonal jump operators $\{J\}$, pairwise dephasing rates satisfy $\Gamma_\phi(j,k) = \frac{1}{2}\|v_j - v_k\|^2$ where v_j collects the j-th diagonal entries across jumps; realizing a target Γ_ϕ matrix is therefore a Euclidean embedding problem, solved exactly by classical multidimensional scaling (residual $\leq 3\times10^{-17}$ for $d = $ 2–7). This realizes the measured max-level law exactly—something no linear-frequency-ladder model can do.

Verification. Twenty correctness tests: QFT circuits against dense matrices; complete positivity and trace preservation of every channel (Choi eigenvalues); trajectory-vs-exact agreement; fidelity-estimator cross-check; noiseless baselines; grid-alignment predicates; unbiased-instance orderings; Grover correctness and cost accounting; and readout-channel monotonicity. The exact decoder count of Sec. VII C and the law of Eq. (2) are verified by outcome-for-outcome comparison against decoder enumeration on 27 instance/size combinations. The external reproduction of Ref. [30] (Sec. VIII) runs separately from the test suite. Statistics: 1000 trajectories per Shor and QPE scaling point (500 at d=2, m=12); 400 per point for the $N = 29$ instance-robustness sweep (200 at the three deepest configurations, whose work registers are wider than at $N = 21$); 400 trajectories per point for the grid-alignment and within-modulus studies; 1000 for the eigenstate-interpolation study and for the $d = 7$ demo grid; 200 for the fidelity-collapse Shor points (100 at d=2, m=12), whose Grover points use exact density-matrix evolution; demo-size cost, SPAM, and refocusing results use exact density-matrix evolution throughout, as does the multiplicative-group ensemble over all eleven units of $\mathbb{Z}_{21}^\times$ (Sec. III); its companions over all nineteen units of $\mathbb{Z}_{33}^\times$ and all thirty-nine of $\mathbb{Z}_{55}^\times$ use 400 trajectories per point. Error bars are standard errors over trajectories; cross-base differences quoted with uncertainties are propagated through the interpolation onto the common size axis.

Hardware runs. Via AWS Braket (August 2026). IonQ Forte-1: 1,000-shot raw pilots, a 500-shot $m = 4$ probe, and 5,000-shot main runs with IonQ debiasing; IonQ's native ZZ accepts $|\theta| \leq \pi/2$, so each controlled-phase compiles as $R_z(\theta/2) \otimes R_z(\theta/2) \cdot ZZ(-\theta/2)$ with any excess folded into paired $R_z(\pi)$ rotations—every compiled variant reproduces the analytic outcome distribution on the Braket local simulator to 10^{-15} before submission. IQM Garnet: 5,000 shots per circuit, server-side routing, native controlled-phase support. Success windows, floors, and the digit-reversed scoring are identical to the simulation pipeline; per-task records and scoring code ship with the repository.

Reproducibility. All code, calibration fits, result files, and the exact scripts that produce every figure and table are available at `https://github.com/doxaras/qudit-decoherence`; the release snapshot corresponding to this paper is archived at doi:10.5281/zenodo.21901534.

Appendix A: The decoder acceptance lemma

The decoder used throughout this paper is the following procedure. On outcome $y \in [1, D)$, scan the continued-fraction convergents p/q of y/D in order of increasing denominator; at the first $q \leq N$ with $a^q \equiv 1$

(mod N), return the least r' dividing q with $a^{r'} \equiv 1$ (mod N); if no convergent denominator $q \leq N$ passes the test, reject ($y = 0$ is always rejected).

Lemma. The decoder returns the order r of a modulo N if and only if some convergent denominator q of y/D satisfies $q \leq N$ and $r \mid q$; on every other outcome it rejects. In particular it never returns a wrong order.

Proof. Since r is the multiplicative order of a modulo N, $a^q \equiv 1 \pmod N$ if and only if $r \mid q$. ($\Leftarrow$) Suppose some convergent denominator $q \leq N$ is a multiple of r; the scan stops at the first such q. Every divisor r' of q with $a^{r'} \equiv 1$ is itself a multiple of r, and r both divides q and passes the test, so the least such r' is exactly r. ($\Rightarrow$) If no convergent denominator $q \leq N$ is a multiple of r, the test $a^q \equiv 1$ never fires and the decoder rejects. □

The equivalence is verified empirically by `decoder_formula.py` in the repository: enumerating the decoder over every outcome $y \in [0, D)$ and comparing against the convergent predicate gives bit-identical acceptance sets on 42 (instance, D) combinations—six instances ($N = 21$, 29, and the four within-modulus orders of $N = 33, 55$) across seven control dimensions each.

Relation to the divisor form. Standard analyses certify a different event: for y within $1/(2\tilde{r}^2)$ of a peak s/r, the convergent guarantee [33] yields the reduced fraction $(s/g)/(r/g)$ with $g = \gcd(s, r)$, whose denominator $\tilde{r} = r/g$ *divides* r [22, 37]. A divisor with $g > 1$ fails the test $a^{\tilde{r}} \equiv 1$, so those analyses lift $\tilde{r}$ to r by classical search over multiples (or over gcd-smoothness classes [37]). The decoder above performs no lift: it succeeds precisely through the convergent denominators that are already multiples of r—including the deeper convergents $kr \leq N$, $k \geq 2$, outside the sufficient window—which is why its acceptance set obeys the totient sum of Eq. (2) over the admissible denominators $r, 2r, \ldots, \lfloor N/r \rfloor r$ rather than a window count.

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[1] M. Ringbauer, M. Meth, L. Postler, R. Stricker, R. Blatt, P. Schindler, and T. Monz, A universal qudit quantum processor with trapped ions, Nat. Phys. **18**, 1053 (2022), arXiv:2109.06903.

[2] M. S. Blok, V. V. Ramasesh, T. Schuster, K. O'Brien, J. M. Kreikebaum, D. Dahlen, A. Morvan, B. Yoshida, N. Y. Yao, and I. Siddiqi, Quantum information scrambling on a superconducting qutrit processor, Phys. Rev. X **11**, 021010 (2021), arXiv:2003.03307.

[3] N. Goss, A. Morvan, B. Marinelli, B. K. Mitchell, L. B. Nguyen, R. K. Naik, L. Chen, C. Jünger, J. M. Kreikebaum, D. I. Santiago, J. J. Wallman, and I. Siddiqi, High-fidelity qutrit entangling gates for superconducting circuits, Nat. Commun. **13**, 7481 (2022), arXiv:2206.07216.

[4] P. J. Low, B. White, and C. Senko, Control and readout of a 13-level trapped ion qudit, arXiv:2306.03340 (2023), arXiv:2306.03340.

[5] A. Gardill, M. C. Cambria, and S. Kolkowitz, Fast relaxation on qutrit transitions of nitrogen-vacancy centers in nanodiamonds, Phys. Rev. Appl. **13**, 034010 (2020), arXiv:1910.10813.

[6] A. Chiesa, E. Macaluso, and S. Carretta, Quantum information processing with molecular nanomagnets: an introduction, arXiv:2405.21000 (2024), arXiv:2405.21000.

[7] D. Gottesman, Fault-tolerant quantum computation with higher-dimensional systems, Chaos, Solitons & Fractals **10**, 1749 (1999), arXiv:quant-ph/9802007.

[8] D. Gross, Hudson's theorem for finite-dimensional quantum systems, J. Math. Phys. **47**, 122107 (2006), arXiv:quant-ph/0602001.

[9] E. T. Campbell, H. Anwar, and D. E. Browne, Magic-state distillation in all prime dimensions using quantum Reed–Muller codes, Phys. Rev. X **2**, 041021 (2012), arXiv:1205.3104.

[10] E. T. Campbell, Enhanced fault-tolerant quantum computing in d-level systems, Phys. Rev. Lett. **113**, 230501 (2014), arXiv:1406.3055.

[11] J. Marks, T. Jochym-O'Connor, and V. Gheorghiu, Comparison of memory thresholds for planar qudit geometries, New J. Phys. **19**, 113022 (2017), arXiv:1701.02335.

[12] A. Barenco, A. Ekert, K.-A. Suominen, and P. Törmä, Approximate quantum Fourier transform and decoherence, Phys. Rev. A **54**, 139 (1996).

[13] Y. S. Nam and R. Blümel, Performance scaling of Shor's algorithm with a banded quantum Fourier transform, Phys. Rev. A **86**, 044303 (2012).

[14] A. Pavlidis and E. Floratos, Quantum-Fourier-transform-based quantum arithmetic with qudits, Phys. Rev. A **103**, 032417 (2021), preprint titled "Arithmetic circuits for multilevel qudits based on quantum Fourier transform", arXiv:1707.08834.

[15] E. O. Kiktenko, A. S. Nikolaeva, and A. K. Fedorov, Colloquium: Qudits for decomposing multiqubit gates and realizing quantum algorithms, Rev. Mod. Phys. **97**, 021003 (2025), arXiv:2311.12003.

[16] A. S. Nikolaeva, E. O. Kiktenko, and A. K. Fedorov, Efficient realization of quantum algorithms with qudits, EPJ Quantum Technol. **11**, 43 (2024), arXiv:2111.04384.

[17] A. Bocharov, M. Roetteler, and K. M. Svore, Factoring with qutrits: Shor's algorithm on ternary and metaplectic quantum architectures, Phys. Rev. A **96**, 012306 (2017), arXiv:1605.02756.

[18] Y. Wang, Z. Hu, B. C. Sanders, and S. Kais, Qudits and high-dimensional quantum computing, Front. Phys. **8**, 589504 (2020), arXiv:2008.00959.

[19] P. Gokhale, J. M. Baker, C. Duckering, N. C. Brown, K. R. Brown, and F. T. Chong, Asymptotic improvements to quantum circuits via qutrits, in *Proc. 46th Int.*

Symp. on Computer Architecture (ISCA) (2019) p. 554, arXiv:1905.10481.

[20] E. Gustafson, Noise improvements in quantum simulations of sQED using qutrits, arXiv:2201.04546 (2022), arXiv:2201.04546.

[21] V. Parasa and M. Perkowski, Quantum phase estimation using multiple-valued logic, in *Proc. 41st IEEE Int. Symp. on Multiple-Valued Logic (ISMVL)* (2011) p. 224, dOI 10.1109/ISMVL.2011.47.

[22] P. W. Shor, Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer, SIAM J. Comput. **26**, 1484 (1997), arXiv:quant-ph/9508027.

[23] A. Y. Kitaev, Quantum measurements and the Abelian stabilizer problem, arXiv:quant-ph/9511026 (1995), arXiv:quant-ph/9511026.

[24] L. K. Grover, A fast quantum mechanical algorithm for database search, in *Proc. 28th ACM Symp. on Theory of Computing (STOC)* (1996) p. 212, arXiv:quant-ph/9605043.

[25] M. J. Peterer, S. J. Bader, X. Jin, F. Yan, A. Kamal, T. Gudmundsen, P. J. Leek, T. P. Orlando, W. D. Oliver, and S. Gustavsson, Coherence and decay of higher energy levels of a superconducting transmon qubit, Phys. Rev. Lett. **114**, 010501 (2015), arXiv:1409.6031.

[26] V. Tripathi, N. Goss, A. Vezvaee, L. B. Nguyen, I. Siddiqi, and D. A. Lidar, Qudit dynamical decoupling on a superconducting quantum processor, Phys. Rev. Lett. **134**, 050601 (2025), arXiv:2407.04893.

[27] Z. Wang, R. W. Parker, E. Champion, and M. S. Blok, Systematic study of high e_j/e_c transmon qudits up to $d = 12$, arXiv:2407.17407 (2024), arXiv:2407.17407.

[28] J. Keppens, Q. Eggerickx, V. Levajac, G. Simion, and B. Sorée, Qudit vs. qubit: Simulated performance of error correction codes in higher dimensions, arXiv:2502.05992 (2025), arXiv:2502.05992.

[29] E. J. Gustafson, H. Lamm, D. Liu, E. M. Murairi, and S. Zhu, Synthesis of single qutrit circuits from Clifford+R, arXiv:2503.20203 (2025), arXiv:2503.20203.

[30] D. Janković, J.-G. Hartmann, M. Ruben, and P.-A. Hervieux, Noisy qudit vs multiple qubits: conditions on gate efficiency for enhancing fidelity, npj Quantum Inf. **10**, 59 (2024), arXiv:2302.04543.

[31] M. A. Yurtalan, J. Shi, G. J. K. Flatt, and A. Lupascu, Characterization of multi-level dynamics and decoherence in a high-anharmonicity capacitively shunted flux circuit, arXiv:2008.00593 (2020), arXiv:2008.00593.

[32] D. Venturelli, E. Gustafson, D. Kurkcuoglu, and S. Zorzetti, Near-term application engineering challenges in emerging superconducting qudit processors, arXiv:2506.05608 (2025), arXiv:2506.05608.

[33] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed. (Oxford University Press, 2008).

[34] A. Y. Khinchin, *Continued Fractions* (University of Chicago Press, Chicago, 1964).

[35] E. Gerjuoy, Shor's factoring algorithm and modern cryptography. An illustration of the capabilities inherent in quantum computers, Am. J. Phys. **73**, 521 (2005).

[36] P. S. Bourdon and H. T. Williams, Sharp probability estimates for Shor's order-finding algorithm, Quantum Inf. Comput. **7**, 522 (2007), arXiv:quant-ph/0607148.

[37] M. Ekerå, On the success probability of quantum order finding, ACM Trans. Quantum Comput. **5**, 11 (2024), arXiv:2201.07791.

[38] M. Magdon-Ismail and K. Dong, Tight success probabilities for quantum period finding and phase estimation, arXiv:2506.20527 (2025), arXiv:2506.20527.

[39] J. Barzen and F. Leymann, Continued fractions and probability estimations in Shor's algorithm: A detailed and self-contained treatise, AppliedMath **2**, 393 (2022).

[40] J. M. Chappell, M. A. Lohe, L. von Smekal, A. Iqbal, and D. Abbott, A precise error bound for quantum phase estimation, PLoS ONE **6**, e19663 (2011).

[41] M. Rubín-Osanz, L. Bersani, S. Chicco, G. Allodi, *et al.*, Implementation of the quantum Fourier transform on a molecular qudit with full refocusing and state tomography, arXiv:2512.15611 (2025), arXiv:2512.15611.

[42] T. Roy, Z. Li, E. Kapit, and D. I. Schuster, Realization of two-qutrit quantum algorithms on a programmable superconducting processor, arXiv:2211.06523 (2022), arXiv:2211.06523.

[43] X. Shi, J. Sinanan-Singh, T. J. Burke, J. Chiaverini, and I. L. Chuang, Efficient implementation of a quantum algorithm with a trapped ion qudit, arXiv:2506.09371 (2025), arXiv:2506.09371.

[44] H.-C. Weng and C.-S. Chuu, Implementation of Shor's algorithm with a single photon in 32 dimensions, arXiv:2408.08138 (2024), arXiv:2408.08138.

[45] H.-H. Lu, Z. Hu, M. S. Alshaykh, A. J. Moore, Y. Wang, P. Imany, A. M. Weiner, and S. Kais, Quantum phase estimation with time-frequency qudits in a single photon, Adv. Quantum Technol. **3**, 1900074 (2020), arXiv:1906.11401.

[46] J. Dalibard, Y. Castin, and K. Mølmer, Wave-function approach to dissipative processes in quantum optics, Phys. Rev. Lett. **68**, 580 (1992).

[47] K. Mølmer, Y. Castin, and J. Dalibard, Monte Carlo wave-function method in quantum optics, J. Opt. Soc. Am. B **10**, 524 (1993).